

LeetCode Bit Manipulation Roadmap

A structured problem list from beginner to advanced. Each problem includes a clickable LeetCode link.

Core concepts to know

AND (&), OR (|), XOR (^), NOT (~), left shift (<<), right shift (>>), $n \& 1$ for odd/even, and $n \& (n-1)$ to remove the rightmost set bit.

1. Beginner — Build Your Bit Manipulation Foundation

#	Problem	Main Concept	LeetCode Link
136	Single Number	XOR	Open Problem
191	Number of 1 Bits	$n \& (n-1)$	Open Problem
338	Counting Bits	DP + Bits	Open Problem
231	Power of Two	$n \& (n-1)$	Open Problem
190	Reverse Bits	Bit shifting	Open Problem
268	Missing Number	XOR	Open Problem
461	Hamming Distance	XOR	Open Problem
476	Number Complement	Masking	Open Problem
693	Binary Number with Alternating Bits	Bit operations	Open Problem
2220	Minimum Bit Flips to Convert Number	XOR + bit checking	Open Problem

2. Intermediate — Apply Bit Tricks

#	Problem	Main Concept	LeetCode Link
371	Sum of Two Integers	XOR + Carry	Open Problem
260	Single Number III	XOR + rightmost set bit	Open Problem
137	Single Number II	Bit counting	Open Problem
201	Bitwise AND of Numbers Range	Common prefix	Open Problem
1318	Minimum Flips to Make a OR b Equal to c	Bit checking	Open Problem
1442	Count Triplets That Can Form Two Arrays of Equal XOR	Prefix XOR	Open Problem
1486	XOR Operation in an Array	XOR	Open Problem
1720	Decode XORed Array	XOR	Open Problem
1734	Decode XORed Permutation	XOR	Open Problem
1863	Sum of All Subset XOR Totals	XOR + subsets	Open Problem

3. Advanced — Bitmasking & Bitmask DP

#	Problem	Main Concept	LeetCode Link
89	Gray Code	XOR	Open Problem

#	Problem	Main Concept	LeetCode Link
78	Subsets	Bitmask	Open Problem
90	Subsets II	Bitmask + Backtracking	Open Problem
464	Can I Win	Bitmask DP	Open Problem
847	Shortest Path Visiting All Nodes	Bitmask DP + BFS	Open Problem
1178	Number of Valid Words for Each Puzzle	Bitmask	Open Problem
1239	Maximum Length of a Concatenated String with Unique Characters	Bitmask	Open Problem
1494	Parallel Courses II	Bitmask DP	Open Problem
1601	Maximum Number of Achievable Transfer Requests	Bitmask	Open Problem
1681	Minimum Incompatibility	Bitmask DP	Open Problem

Recommended solving order

136 → 191 → 231 → 338 → 268 → 461 → 476 → 190 → 2220 → 371 → 260 → 137 → 201 → 1318 → 78 → 90 → 1239 → 464

Tip: Try each problem yourself first, then study the bit trick only when you get stuck.